Do Now · retrieval

Lesson 093 · Von Neumann and Harvard Architectures

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does an address bus carry?

Q2

What does a data bus transfer?

Q3

Define contention.

Q4

A D flip-flop has D=1 and Q=0 immediately before a rising edge. What is Q just after?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — Von Neumann stores programs and their data in a common memory/address space, so instruction fetches and data transfers may contend for shared resources.

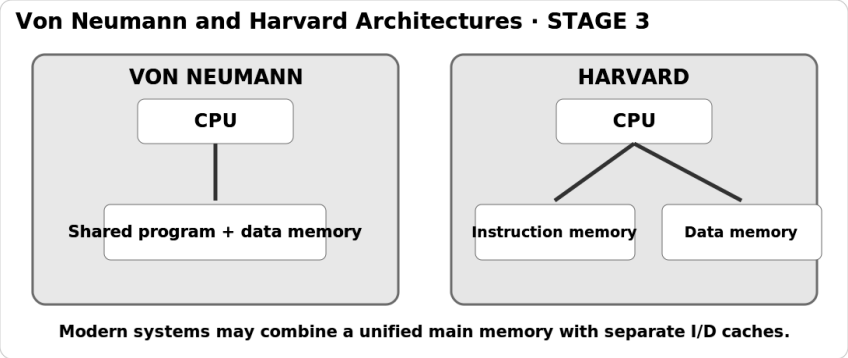
Core idea 2

Explain — Harvard architecture separates instruction and data pathways, potentially allowing simultaneous instruction fetch and data access.

Core idea 3

Evaluate — Modern systems often use unified main memory but separate instruction/data caches; the classification is a model of design trade-offs, not a universal speed ranking.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

Von Neumann — Shared instruction/data memory organisation.

Harvard — Separate instruction/data stores or pathways.

bottleneck — Limiting shared resource.

modified Harvard — Modern mixed memory hierarchy.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

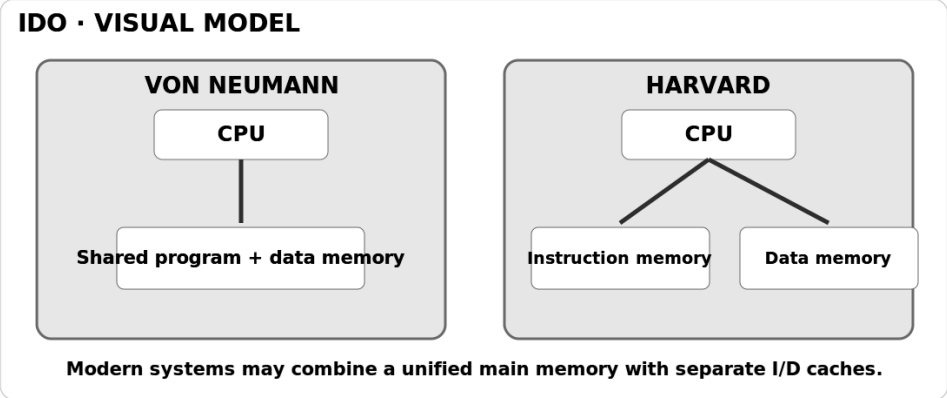
Problem

Explain one potential bottleneck of simple Von Neumann design.

Follow the worked method

1. Explain the mechanism — Instruction and data transfers share memory/bus resources, so competing accesses can wait. — Describe the shared resource and its effect, not claim all Von Neumann machines are slow.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the mechanism
Harvard-style separation; fetch instruction and access data paths may operate in parallel.
It differs from a single shared path even when control logic is the same CPU.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

State difference between Von Neumann and Harvard memory organisation. [4]

Question 2

Explain Von Neumann bottleneck with instruction fetch and data read. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate 'Harvard architecture always makes programs faster than Von Neumann' with contemporary caches and workload. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Contemporary processors can combine a unified main memory with separate instruction and data caches.

Use key vocabulary from memory

1. Explain Von Neumann in your own words.

2. Explain Harvard in your own words.

3. Explain bottleneck in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.